Naive Lie Theory 4.2 Exercises

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Notes

Nothing much to say here.

4.2.1

Say there is a line $l := \{tu | t \in \mathbb{R}\}$ for some fixed unit vector $u \in \mathbb{R}i + \mathbb{R}j + \mathbb{R}k$. Therefore the exponential map restricted to l is just

$$\exp|_l(\theta) = \cos\theta + u \sin\theta$$

The image lies in the plane spanned by $1, u$, and it satisfies that the norm is 1, so it's a circle.

4.2.2

We want $\cos\theta + u\sin\theta = i$, so $u = i$ and $\theta = \frac{\pi}{2}$. Similarly, for $\cos\theta + u\sin\theta = j$, we need $u = j$ and $\theta = \frac{\pi}{2}$.

4.2.3

From 4.2.2 we have $e^{i\frac{\pi}{2}} e^{j\frac{\pi}{2}} = ij$ and $e^{j\frac{\pi}{2}} e^{i\frac{\pi}{2}} = ji$. We know that $ij \neq ji$. Thus they cannot both be $e^{(i+j)\frac{\pi}{2}}$.